

# Tide retrospective: the signed exponential remainder (gamma-rung programme, stage 1 of 6)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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## 1 Setting

This tide opens the gamma-rung programme: completing the anharmonic jet recovery  $(\lambda, \alpha, \gamma)$  of the germbij note’s Theorem 3.1 by proving  $t^3\kappa_4 \rightarrow -\gamma/\lambda^4 + 3\alpha^2/\lambda^5$  and its recovery corollary. The programme was scoped in a dedicated consult before any Lean: the target was pinned numerically first (high-precision quadrature with a spike breakpoint, converging to the candidate at four digits), the full symbolic route was verified (second-order  $J_n$  coefficients, moment ratios, and the fourth-cumulant cancellation table summing to  $108A^2 - 24B = 3\alpha^2/\lambda^5 - \gamma/\lambda^4$ ), the staging fixed at six tides, and the named risks checked against the seabed (the coercivity risk is already discharged by the standing discriminant hypothesis). Stage 1 is the smallest self-contained rung: the rescaled anharmonic perturbation  $s_t(u)$  is sign-indefinite, so the second-order expansions need the exponential Taylor remainder for *signed* arguments.

## 2 The thing formalised

Two theorems in `Laplace/OneD/ExpRemainderSigned.lean`, sorry-free with zero warnings. The sharp nonpositive-axis form<sup>1</sup>: for  $s \leq 0$ ,  $|E_n(s)| \leq |s|^n/n! \cdot e^{-s}$ . And the two-sided endpoint-maximum form<sup>2</sup>: for every real  $s$ ,

$$\left| e^{-s} - \sum_{j < n} \frac{(-s)^j}{j!} \right| \leq \frac{|s|^n}{n!} \max(1, e^{-s}),$$

the form the scoping consult specified as composing with the seabed’s existing Gaussian-domination layer (whose first-order `ScalarBound` has exactly this two-branch shape).

## 3 Proof strategy

The same fundamental-theorem induction as the nonnegative case, run on  $[s, 0]$ : the previous remainder at  $u \in [s, 0]$  carries the sharp bound with  $e^{-u} \leq e^{-s}$ , and the polynomial integral evaluates by reflection (`integral_comp_neg` then the power rule). The sharp  $e^{-s}$  form is what makes the induction close — the max only enters in the final packaging, where the nonnegative branch reuses the merged all-order bound against  $\max \geq 1$ .

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<sup>1</sup>`abs_expRemainder_le_of_nonpos`.

<sup>2</sup>`abs_expRemainder_le_max`.

## 4 Roadblocks and resolutions

Three build iterations, all familiar classes: an integral-orientation identity is stated as its own `have` (`integral_symm` plus `abs_neg`) rather than inlined in a rewrite chain; `positivity` cannot derive  $(-u)^n \geq 0$  from a hypothesis  $u \leq 0$  (explicit `pow_nonneg`); `le_or_lt` no longer exists (`le_total`); and a `field_simp` closed a goal leaving a `ring` dangling.

## 5 What was Mathlib, what was new

Mathlib: interval integration, the reflection substitution, and the power rule. Seabed: the exponential remainder recursion of the recovery thread,<sup>3</sup> which this file extends. New: the observation that the sharp  $e^{-s}$  inductive form is the right strengthening for the signed axis.

## 6 Lessons

When an induction needs a factor that grows along the integration interval, put the endpoint value in the inductive statement (here  $e^{-s}$ ), not the pointwise one; the packaging into the user-facing `max` form is then a two-line case split. Programme scoping with a numerically pinned target and a symbolic verification before stage 1 meant this tide began with zero mathematical uncertainty.

## 7 Follow-ups

- Stage 2: second-order  $J_n$  asymptotics ( $J_n = m_n - Am_{n+3}t^{-1/2} + (A^2/2 m_{n+6} - Bm_{n+4})t^{-1} + O(t^{-3/2})$ ), consuming this tide's bound at  $n = 3$  against the seabed's Gaussian domination.
- Stages 3–6 per the programme plan (moment ratios,  $\kappa_4$  assembly, the analytic limit, gamma recovery).

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<sup>3</sup>`expRemainder` and its derivative and endpoint lemmas, reused without change.